Note 3 - Transformations of Functions

Transformations

- **Four Shifts**
  - Up by c units: $f(x) + c$
  - Down by c units: $f(x) - c$
  - Right by c units: $f(x - c)$
  - Left by c units: $f(x + c)$
- **Two Reflections**
  - Through x-axis: $-f(x)$
  - Through y-axis: $f(-x)$
- **Vertical Stretch/Shrink**
  - Stretch by a factor of c units: $cf(x)$ where $c > 1$
  - Shrink by a factor of $\frac{1}{c}$ units: $\frac{1}{c}(x)$ where $c > 1$
    Recall: if $c < 0$, then it is a reflection combined with a stretch/shrink!
- **Horizontal Stretch/Shrink**
  - Stretch by a factor of c units: $f(\frac{x}{c})$ where $c > 1$
  - Shrink by a factor of c units: $f(cx)$ where $c > 1$
- **Mother Functions**: Base functions that are often transformed.
  - $y = |x|$
  - $y = \sqrt{x}$
  - $y = x^2$
  - $y = x^3$
- **Order of Operations**: Given a Mother Function, perform transformations in this order:
  1. All stretches or shrinks
  2. All reflections
  3. All shifts

Composite Functions

- f composed with g: $(f \circ g)(x) = f(g(x))$
- g composed with f: $(g \circ f)(x) = g(f(x))$
- f composed with f: $(f \circ f)(x) = f(f(x))$
- g composed with g: $(g \circ g)(x) = g(g(x))$

For composite functions, evaluate the inner function first, with respect to the input x. The output of the inner function, $g(x)$, becomes the input of the outer function f *(for the first case)*. If we set $y = g(x)$, then the final answer is the output after evaluating $f(y)$.